

Djordje Batic

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AI Research Engineer with 5+ years of experience designing and evaluating AI systems across finance and energy. Currently working on **AI Safety** at Lloyds Banking Group (risk, governance, and guardrails for generative and agentic AI). Previously PhD in Trustworthy AI (Marie Skłodowska-Curie Fellow) with expertise in model training and evaluation, interpretability, efficient edge deployment, and deep domain knowledge in the energy sector.

Education

- **Ph.D. in Electronic & Electrical Engineering - University of Strathclyde** Glasgow, UK
Focus: Trustworthy AI at the Intersection of AI and Energy Research | Marie Curie Fellow Oct 2021 - May 2025
- **M.Sc. in Electrical & Computer Engineering - University of Novi Sad** Novi Sad, Serbia
Focus: Computer Science, ML, Robotics, Control Systems, Automation | CGPA: 9.50/10.00 Sep 2016 - Sep 2021

Work Experience

- **Lloyds Banking Group** Edinburgh, UK
Senior AI & Data Scientist | AI Safety 2026 – Present
 - **Low-Latency LLM Guardrails:** Reduced median latency by 20% and tail latency by 90% through architectural changes to the guardrails deployed within the bank's central LLM gateway.
 - **Guardrail Cost Optimization:** Reduced the cost of running guardrails by over 40% by developing a context caching solution for LLM inference.
 - **LLM Emissions Estimation & Sustainability Reporting:** Developed an approach for estimating carbon emissions from LLM API usage across all genAI and agentic use cases served through the bank's gateway, adopted for bank-wide sustainability reporting and auditing.
- **University of Strathclyde** Glasgow, UK
Marie Curie Early Stage Researcher (2021-2025) and Postdoctoral Researcher (2025) Oct 2021 – Aug 2025
 - **Explainable Knowledge Distillation for Edge AI [1], [4]:** Developed an explainability-guided knowledge distillation method for edge deployment of energy disaggregation models, improving predictive performance by up to 22.6% at low computational overhead.
 - **Trustworthy AI Evaluation & Training [2], [3]:** Proposed a framework for evaluating trustworthiness of (X)AI systems with metrics for robustness, faithfulness, and sparsity; developed a model-agnostic training approach using attribution priors, jointly improving interpretability and predictive performance across architectures.
 - **GNNs for Spatio-Temporal Systems [5], [6]:** Applied GNNs to optimal EV charging infrastructure placement, combining graph embeddings with greedy search for scalable planning, improving simulated charging utilization by up to 55%.
 - **Large-Scale Datasets [7]:** Created and open-sourced GridCharge, a dataset of 5M+ public EV charging sessions in Scotland with carbon intensity, occupancy, and weather data, leveraging LLMs to structure thousands of unstructured tariff documents. | [Github](#) | [HuggingFace](#)
 - **Consultancy:** PictureIT - built an on-device speech-to-image app generating AAC-style images for education of neurodivergent children (fine-tuned diffusion adapter), securing two rounds of government funding; Hugo Energy - led energy-flexibility profiling of 50,000+ UK smart-meter customers in partnership with University of Oxford and University of East Anglia, creating behavioral segments for demand-response programs; Flock Energy - developed an energy disaggregation training pipeline delivering usage insights to customers in Asian markets. | [Github](#)
- **BioSense Institute** Novi Sad, Serbia
Research Assistant Jan 2021 – Oct 2021
 - Developed a multi-task learning approach for joint boundary delineation and instance segmentation of agricultural parcels from satellite imagery. Findings led to adoption of new multi-task pipeline to replace existing single-task baselines for crop monitoring.
- **Cinteraction**
AI Development Consultant Aug 2020 – Nov 2020
 - Designed and optimized a CNN model for real-time facial affect analysis, leveraging CUDA inference optimization using double-buffering. Deployed on mobile devices, achieving 57% improvement in throughput. | [Github](#)

SKILLS

- Python, Pytorch, JAX, C/C++, CUDA, Java, SQL, Git, Docker, Slurm, AWS, GCP, L^AT_EX

TEACHING AND MENTORING

- Engineering Design For Software Development
- Image And Video Processing

AWARDS AND FUNDING

- **2021:** Marie Curie Doctoral Fellowship (~€125,000) - EU Horizon 2020 MSCA-ITN Program
- **2023-2025:** Secured over £20k from consultancy projects, contributed to additional ~£200k funding from proposals.

LANGUAGES

- English (Fluent), Serbo-Croatian (Native), German (Basic)

SELECTED PUBLICATIONS

- [1] **Batic, D.**, Tanoni, G., Stankovic, L., Stankovic, V., Principi, E., “[Improving Knowledge Distillation for Non-Intrusive Load Monitoring through Explainability Guided Learning](#)”, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [2] **Batic, D.**, Stankovic, V., Stankovic, L., “[XNILMBoost: Explainability-informed Load Disaggregation Training Enhancement using Attribution Priors](#)”, *Engineering Applications of Artificial Intelligence*, 2024.
- [3] **Batic, D.**, Stankovic, V., Stankovic, L., “[Towards Trustworthy Load Disaggregation – A Framework for Quantitative Evaluation of Explainability using Explainable AI](#)”, *IEEE Transactions on Consumer Electronics*, 2023.
- [4] **Batic, D.**, Tanoni, G., Principi, E., Stankovic, L., Stankovic, V., Squartini, S., “[Interpretability and Reliability-driven Knowledge Distillation for Non-intrusive Load Monitoring on the Edge](#)”, *Expert Systems with Applications*, 2025.
- [5] **Batic, D.**, Stankovic, V., Stankovic, L., “[Geodemographic Aware Electric Vehicle Charging Location Planning for Equitable Placement Using Graph Neural Networks: Case Study of Scotland Metropolitan Areas](#)”, *Energy*, 2025.
- [6] **Batic, D.**, Stankovic, V., Stankovic, L., “[ChargeDEM: Geodemographic Aware EV Charging Infrastructure Placement for Enhanced Site Selection using Graph Neural Networks](#)”, *International Conference on Energy Efficiency in Domestic Appliances and Lighting (EEDAL)*, 2024.
- [7] **Batic, D.**, “[GridCharge: Capturing Carbon Intensity of 5M+ Public EV Charging Sessions in Scotland](#)”, 2025.
- [8] **Batic, D.**, Stankovic, L., Stankovic, V., “[Analysis of Smart Meter Data for Energy Waste Management](#)”, *Artificial Intelligence for Sustainability: Innovations in Business and Financial Services*, 2024.
- [9] **Batic, D.**, “[Towards Trustworthy AI Systems for Smart Grid Management: Facilitating Robustness, Transparency and Fairness in the Energy Transition](#)”, *PhD Thesis*, University of Strathclyde, 2025.
- [10] Ilic, V., **Batic, D.**, Mirkovic, M., Vukmirovic S., Culibrk, D., Bosakov, G., “[Automatic Emotion Detection as a Teaching Aid in Online Knowledge Assessment](#)”, *2021 20th International Symposium INFOTEH*, 2021.
- [11] **Batic, D.**, Culibrk, D., “[Identifying Individual Dogs in Social Media Images](#)”, *2019 British Machine Vision Conference (BMVC) Workshop on Visual AI and Entrepreneurship (VAIE)*, 2019.